Probabilistic Cellular Automaton for the SIR Model

Justin Deckers
UCSC CSE-185

Abstract*—*Epidemic modeling is key to understanding and forecasting disease spread. Classically doing so is done by differential equation-based models like the SIR (Susceptible-Infected-Recovered) model. Although providing valuable insights, these models are based on assumptions of homogeneous population mixing, which may not accurately reflect real-world disease transmission. This paper explores an alternative approach by applying probabilistic cellular automata (PCA) to simulate the SIR model. Representing individuals on a structured grid and applying statistical transition rules, this method simulates dynamic epidemic spreading both on the individual and spatial level. We will analyze the impact of infection and recovery probabilities on epidemic dynamics, contrast resulting patterns with classic models, and test interventions like social distancing and vaccinations. Our results will show how probabilistic cellular automata can provide a more realistic and modular system for the study of infectious disease.

Keywords—Probabilistic cellular automata, epidemic modeling, SIR model, spatial epidemiology, stochastic simulation, disease spread, infection dynamics, epidemic thresholds

I. Introduction

1.1 Overview of the SIR Model

The Susceptible-Infected-Recovered (SIR) model is a fundamental framework in epidemiology, used to describe the spread of infectious diseases within a population. It categorizes individuals into three compartments:

- **Susceptible (S)**: Individuals who can contract the disease.
- **Infected (I)**: Individuals who have the disease and can spread it.
- **Recovered (R)**: Individuals who have recovered from the disease.

The standard SIR model is represented by a system of ordinary differential equations (ODEs) that models the rate of change in each population group over time. While suitable for general prediction, ODE-based models assume homogeneous mixing, i.e., all individuals have an equal chance of encountering all other individuals. In reality, however, disease transmission is dependent on spatial organization, individual contact, and probabilistic events.

1.2 Why Cellular Automata?

Cellular automata (CA) offers an individual, spatially explicit alternative to traditional models. In a probabilistic cellular automaton (PCA), individuals are mapped to a discrete grid and only interact with neighbors, simulating real contact. This allows us to study local transmission patterns and emergent behavior that would be lost in differential equation models. Key advantages of this approach are:

- **Spatial dynamics**: Infections spread based on local interactions rather than a mixed population.
- **Modeling method**: Disease transmission depends on neighborhood structure rather than uniform probability.
- **Modular probability:** The randomness inherent in real epidemics is naturally incorporated and can be manipulated to test different values and strategies(e.g. social distancing, vaccinations).

This paper presents a probabilistic cellular automaton SIR model, with a description of its rules, parameters, and probabilistic dynamics. We will explore the effects of changing infection and recovery probabilities on the epidemic model, contrast outcomes and methods with standard models, and analyze the implications of social distancing and vaccination measures. Our findings will illustrate how PCA models can provide more insight into epidemic behavior in scenarios where spatial structure and randomness are key factors.

II. Cellular Automaton Model

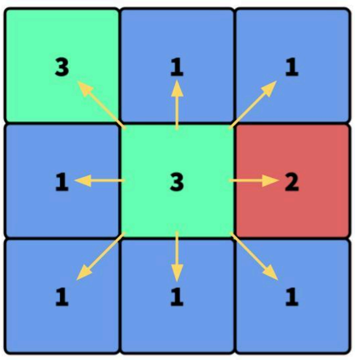
2.1 Grid and Cell Representation

Probabilistic cellular automaton (PCA) for the SIR model is implemented on a two-dimensional grid, where each cell represents an individual. At any given time step, a cell can be in one of three possible states:

- Susceptible (S): The individual is healthy but at risk of infection.
- Infected (I): The individual has contracted the disease and can spread it.
- Recovered (R): The individual has recovered and (in this representation) is assumed immune from further infection.

This discrete spatial representation allows for localized disease transmission, enabling a more realistic simulation of epidemic spread compared to traditional compartmental models.

2.2 Neighborhood Definition



Each cell interacts with its eight adjacent cells surrounding it. This choice allows for a more dynamic spread of infection, as it accounts for interactions from multiple neighbors, including diagonally. The neighborhood configuration influences the rate and structure of disease transmission across the population/grid.

2.3 Transition Rules (Probability)

The state of each cell evolves based on statistical transition rules, reflecting the probabilistic nature of disease transmission and recovery. At each discrete time step, the following transitions occur:

1. Susceptible → Infected (S → I):
   - A susceptible individual becomes infected with probability P_infect.
   - This rule models local transmission by requiring physical proximity for disease spread.
2. Infected → Recovered (I → R):
   - An infected individual recovers with probability P_recover per time step.
   - Once recovered, the individual remains immune and does not return to the susceptible state.

These rules define the epidemic dynamics in the cellular automaton, capturing the statistical and spatial aspects of disease spread. The model's probabilistic nature introduces randomness, allowing for the emergence of infectious outbreak scenarios depending on initial conditions and parameter values.

III. Parameters and Statistical Behavior

The behavior of probabilistic cellular automaton SIR model is governed by two primary parameters that define the progression of the epidemic:

$$P_{\text{infect}} = \frac{1}{1+e^{-(a+bx)}}$$

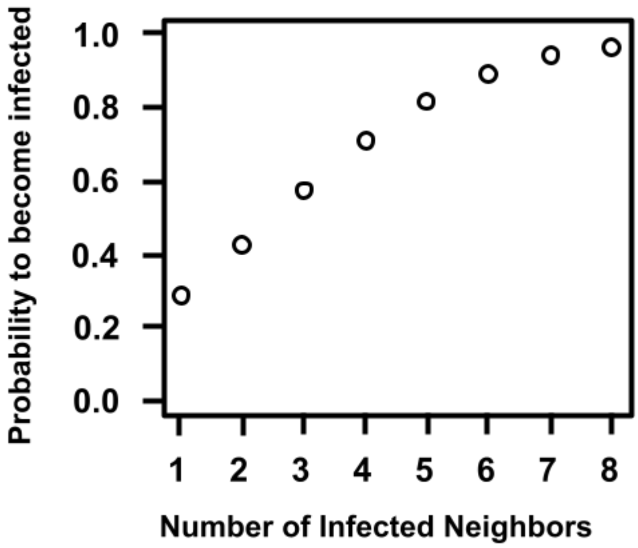
a = -1.5

b = 0.6 **x = Number of Infected Neighbors**

1. Infection Probability (P_infect):
   - Represents the likelihood that a susceptible individual becomes infected when exposed to one or more infected neighbors.
   - Higher values of P_infect lead to more rapid disease spread, potentially causing widespread outbreaks.
   - Lower values may result in slower transmission, localized infections, or even outbreak extinction.

$$P_{\text{recover}} = 0.5$$

2. Recovery Probability (P_recover):
   - Determines the likelihood that an infected individual transitions to the recovered state at each time step.
   - Higher values of P_recover shorten the duration of infection, reducing the number of infected individuals at any time.
   - If P_recover is too low, infections persist for longer, potentially overwhelming the susceptible population.



Both probabilities introduce statistical variability into the system. As we can see above, increasing the number of infected neighbors introduces a higher probability of the individual to become infected.

3.1 Role of Randomness in Modeling Real-World Epidemics

Unlike deterministic models, where infection and recovery rates follow smooth differential equations, the probabilistic nature of cellular automata captures the unpredictability of real-world epidemics. Factors such as:

- Superspreading events (a few individuals infecting disproportionately more neighbors).
- Localized immunity pockets (clusters of recovered individuals limiting further spread).
- Statistical extinction (early infections dying out by chance).

These aspects are naturally emergent in probabilistic simulations, making the model more adaptable to real epidemiological scenarios.

3.2 Emergent Patterns in Disease Transmission

The effect between P_infect and P_recover can lead to different epidemic dynamics, including:

- Endemic spread: If infection persists without complete recovery, leading to sustained but controlled outbreaks.
- Epidemic waves: Infection spreads in waves due to local clusters of susceptible individuals.
- Percolation threshold: If P_infect is below a critical value, the disease may fail to spread beyond a small cluster.

- Herd immunity formation: If a significant portion of the grid reaches the recovered state, the epidemic dies out naturally.

By adjusting parameters governing our model, we can observe transitions where small changes in P_infect or P_recover drastically alter disease dynamics, similar to percolation theory in statistical physics[3].

IV. Implementation / Methodology

4.1 Initialization

The probabilistic cellular automaton for the SIR model begins with an initial state assignment on a 2D grid, with each cell representing an individual. The initialization process determines how the epidemic starts and influences its transmission.

4.1.1 Grid Setup

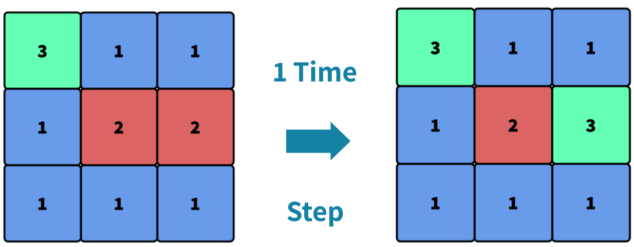
- The grid consists of N × N cells, with each cell in one of three states: Susceptible (S), Infected (I), or Recovered (R).
- The initial infection is assigned in one of two ways:
    1. Random Initialization: A small percentage of randomly chosen cells are set to the infected state (I), while all others remain susceptible (S).
    2. Clustered Initialization: The infection starts in a localized region of the grid, mimicking a geographically concentrated outbreak.

The choice between these two configurations affects the spatial distribution of the disease and can be used to model different real world scenarios.

4.2 Time Evolution

The disease progresses over discrete time steps, with each step updating the state of every cell based on transition probabilities. The update rules follow:

1. Infection (S → I):
    - A susceptible (S) individual becomes infected (I) with probability P_infect if at least one of its 8 neighbors is infected.
2. Recovery (I → R):
    - An infected (I) individual transitions to the recovered (R) state with probability P_recover per time step.



4.2.1 Iterative Update Process

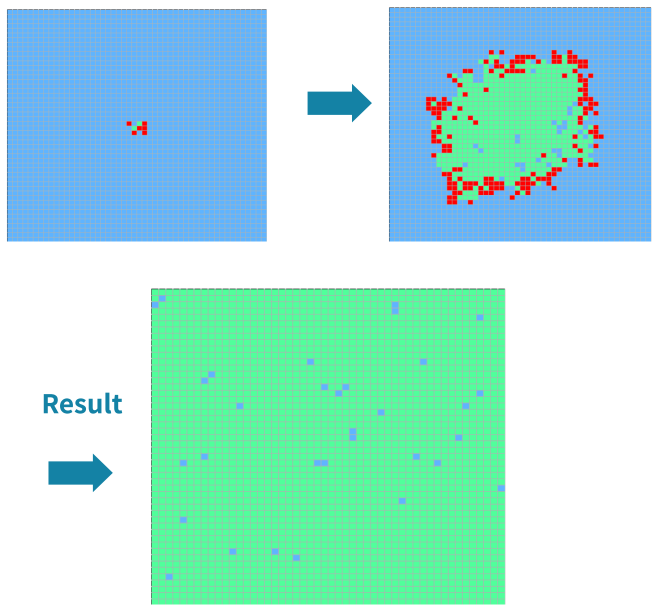
- Each time step, all cells update simultaneously following the above rules.
- The process continues until the epidemic naturally dies out (i.e., no remaining infected individuals) or reaches a steady-state (where infections persist indefinitely in an endemic fashion).

4.3 Visualization and Tracking

To analyze the epidemic progression, the following visualization techniques are employed:

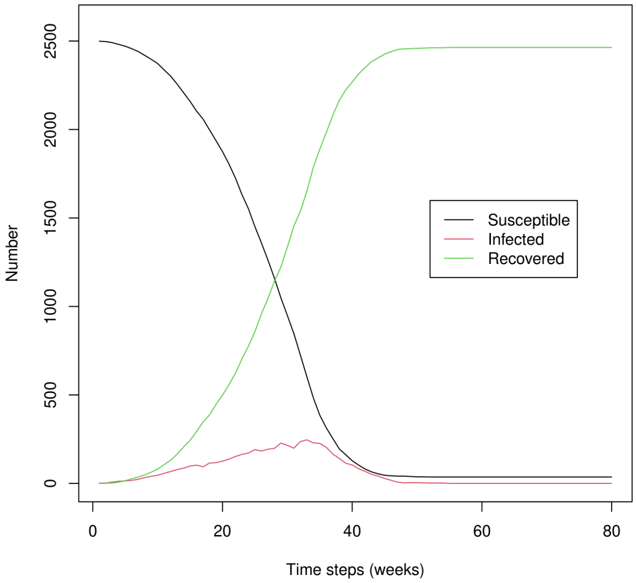
1. Heatmaps of Infection Spread Over Time

- The 2D grid is color-coded at each time step to illustrate the spatial spread of infection:
    - Blue (S): Susceptible individuals.
    - Red (I): Infected individuals.
    - Green (R): Recovered individuals.
- The heatmap provides an intuitive view of how the infection propagates and where immunity builds up.



2. Graphs of S, I, R Population Fractions vs. Time

- The number of individuals in each state (S, I, R) is tracked at every time step.
- A time-series graph is plotted, showing:
    - Susceptible (S) curve: Declining as individuals get infected.
    - Infected (I) curve: Rising initially, peaking, then decreasing as individuals recover.
    - Recovered (R) curve: Increasing over time as more individuals gain immunity.



These graphs allow for a direct comparison to the classical SIR differential equation model, highlighting differences introduced by spatial and statistical effects.

V. Experimental Simulations

To evaluate the behavior of the probabilistic cellular automaton SIR model, we conduct several simulations under different conditions. These experiments explore how the infection probability (P_infect) and recovery probability (P_recover) influence epidemic progression and examine possible interventions such as social distancing and vaccination.

5.1 Baseline Scenario: Uncontrolled Epidemic Spread

In the baseline simulation, the model runs without any interventions, following the standard transition rules:

- A small percentage of the grid is initially infected.
- Disease spreads based on P_infect, and individuals recover based on P_recover.
- The simulation continues until no infected individuals remain.

This scenario establishes a control case to compare the effects of interventions and parameter changes.

Key Observations in the Baseline Scenario

- If P_infect is high and P_recover is low, the epidemic spreads rapidly, infecting most of the population before burning out.
- If P_infect is low and P_recover is high, infections may die out early, preventing widespread outbreaks.
- Spatial effects cause localized outbreaks that may persist longer in some regions while others recover faster.

5.2 Varying Parameters: Effects on Epidemic Dynamics

To understand the impact of statistic variation, we systematically adjust P_infect and P_recover and analyze their effects on:

- Peak infection levels (maximum percentage of individuals infected at once).
- Epidemic duration (how long the infection persists before extinction).
- Final population states (how many individuals remain susceptible or recovered).

Effects of Changing P_infect

- High P_infect (~0.9): The disease spreads rapidly, overwhelming most of the population before recovery kicks in.
- Low P_infect (~0.2): Many susceptible individuals avoid infection, and the disease may die out before reaching a large portion of the population.
- Threshold Behavior: A critical P_infect threshold exists where the infection either dies out early or grows exponentially, similar to percolation theory[3].

Effects of Changing P_recover

- High P_recover (~0.8): The infection is short-lived, as individuals recover quickly before spreading it to many neighbors.
- Low P_recover (~0.1): Infected individuals remain contagious longer, prolonging the epidemic and increasing the total number of cases.
- Delayed Peaks: When P_recover is low, infections persist longer, leading to drawn-out epidemics with prolonged infection waves.

5.3 Intervention Scenarios

To model epidemic control strategies, we introduce interventions and assess their impact on disease spread.

1. Social Distancing: Reduced Neighborhood Size

- Instead of using a Moore neighborhood (8 neighbors), we reduce interactions by using a Von Neumann neighborhood (4 neighbors) or even nearest-neighbor only (2 neighbors).
- Effect:
  - Reduces transmission opportunities.
  - Leads to localized outbreaks rather than widespread epidemics.
  - In extreme cases, infections may fail to transmit through the population, leading to natural containment.

2. Vaccination: Direct S → R Conversion

- A fixed percentage (V%) of the population is converted from Susceptible (S) to Recovered (R) at the beginning of the simulation, modeling preemptive vaccination.
- Effect:
  - If V% is high, herd immunity prevents the epidemic from spreading significantly.

  - If V% is low, the outbreak may still occur but at a reduced scale and slower rate.
  - There exists a critical vaccination threshold where disease spread is effectively blocked.

5.4 Summary of Findings

- Higher P_infect leads to faster and larger outbreaks, while higher P_recover shortens epidemic duration.
- Social distancing reduces infection spread by limiting contact.
- Vaccination prevents outbreaks if a sufficient fraction of the population is immunized.
- The model demonstrates critical thresholds, where small parameter changes can drastically alter epidemic outcomes.

VI. Results and Observations

This section presents the key findings from the probabilistic cellular automaton SIR model simulations. We analyze how infection probability (P_infect) and recovery probability (P_recover) affect epidemic dynamics, compare the PCA model to the classical SIR differential equation model, and discuss critical thresholds that determine epidemic outcomes.

6.1 Emergent Epidemic Dynamics

Spread Patterns in High vs. Low Transmission Probability Settings

The spatial nature of disease transmission in the PCA model results in distinct epidemic patterns based on P_infect and P_recover:

- High P_infect, Low P_recover:
  - The disease spreads rapidly across the grid, with nearly all susceptible individuals becoming infected before recovery occurs.
  - Infection waves emerge, propagating outward from the initial infection cluster.
  - The epidemic reaches a sharp peak before rapidly declining due to population-wide immunity.
- Low P_infect, High P_recover:
  - The infection spreads more slowly and locally, often failing to sustain long-range transmission.
  - Many clusters of susceptible individuals remain uninfected, resulting in a fragmented outbreak with limited propagation.
  - The epidemic may fade out naturally without reaching most of the population.

Comparison with Classic SIR Differential Equation Models

A major distinction between PCA-based models and ODE-based models lies in their handling of spatial effects and stochasticity:

- Classic SIR Model (Differential Equations):
  - Assumes a well-mixed population, meaning every individual has an equal chance of interacting with every other individual.
  - Produces smooth, continuous epidemic curves where the infected population rises and falls predictably.
  - Lacks spatial clustering, making it unable to capture localized outbreaks or containment effects.
- PCA SIR Model:
  - Captures spatial diversity, where disease spread is influenced by local neighborhood structure.
  - Produces statistical variations in epidemic curves, allowing for cases where the epidemic dies out prematurely or spreads in distinct waves.
  - Accounts for localized immunity, where some parts of the grid recover before others, creating natural barriers to disease propagation.

This difference is particularly important in modeling real-world epidemics, where social interactions are not homogeneous and outbreaks often depend on localized transmission.

6.2 Critical Thresholds & Epidemic Evolution

Propagation Behavior in Disease Spread

In probabilistic cellular automata, epidemic transmission is highly dependent on threshold behaviors:

- If P_infect is too low, the disease fails to spread significantly and dies out in the early stages.
- If P_infect crosses a critical value, the infection sustains itself and spreads widely across the grid.
- If P_recover is too high, the infection cannot persist, leading to rapid containment.

This behavior aligns with percolation theory, where the spread of infection resembles a phase transition[3]:

- Below a critical percolation threshold, the disease remains localized and eventually disappears.
- Above the threshold, the infection spreads systemically, reaching a large portion of the population.

Identifying Tipping Points for Outbreak Containment

Through multiple simulation runs, we identify key tipping points where small changes in P_infect or P_recover drastically alter epidemic outcomes:

1. Epidemic Collapse:
   - If P_infect < threshold, early infections fail to sustain, and the epidemic naturally fades out.

- Possible real-world interpretation: Effective social distancing or naturally low transmission rates.

2. Epidemic Expansion:
   - If P_infect exceeds threshold, the infection spreads uncontrollably until most individuals recover.
   - Possible real-world interpretation: Unchecked outbreaks in densely connected populations.
3. Partial Containment:
   - If interventions (e.g., reduced neighborhood interactions or targeted vaccinations) are applied, the outbreak spreads only in pockets, leading to an endemic equilibrium rather than a full outbreak.

6.3 Summary of Observations

- Spatial structure and statistical effects lead to diverse outbreak scenarios, unlike the smooth curves seen in classic SIR models.
- High transmission probabilities result in widespread infections, while low transmission probabilities cause fragmented or failed outbreaks.
- Threshold dynamics play a crucial role, where small changes in parameters can determine whether an epidemic persists or collapses.
- Interventions like social distancing and vaccination create natural barriers that limit disease transmission, mimicking real-world epidemic control strategies.

VII. Future Work

7.1 Summary of Findings

While this study focused on the basic SIR model using probabilistic cellular automata, several extensions can improve the model's applicability and realism:

1. SEIR Model: Adding an Exposed (E) State

- The SEIR model introduces an exposed (E) state, representing individuals who have been infected but are not yet infectious.
- This refinement allows for more realistic incubation periods, where individuals transition:
  - S → E (exposed, but not infectious yet)
  - E → I (becomes infectious after an incubation delay)
  - I → R (recovers as in SIR model)
- The addition of an exposed state would help model diseases like COVID-19, where there is a latency period between infection and symptom onset[2].

2. Incorporating Mobility: Moving Across the Grid

- In the current model, individuals remain stationary, interacting only with their local neighborhood.
- Future models could allow individuals to move, simulating human movement in urban environments, transportation networks, or large gatherings.
- This would introduce disease transmission across distant locations, modeling superspreader events or long-range epidemic waves.

3. More Complex Interventions: Quarantine and Adaptive Measures

- Quarantine Zones:
  - Instead of allowing free movement of infected individuals, quarantining infected cells would simulate lockdown effects.
  - This could be implemented as "restricted movement areas" that block further disease spread[2].
- Dynamic Control Measures:
  - Instead of fixed intervention strategies, future models could incorporate adaptive policies, such as:
    - Triggering stricter social distancing when infection rates exceed a threshold.
    - Applying vaccination rollouts dynamically based on regional infection hotspots.
    - Simulating reactive behaviors, where individuals modify their movement or contact rates based on perceived risk.

VIII. Conclusion

This study explored the probabilistic cellular automaton (PCA) SIR model as an alternative framework for modeling disease spread. Unlike traditional differential equation based SIR models, which assume a mixed population, cellular automata incorporate spatial interactions and statistics, making them more suited for real-world epidemics where disease transmission depends on local interactions and heterogeneous contact patterns.

Key insights from this study include:

- Spatial structure plays a crucial role in epidemic progression. Unlike classical models that assume uniform mixing, the PCA model demonstrates that localized infections, immunity clusters, and spatial barriers can dramatically alter disease dynamics.
- Probabilistic transitions introduce randomness, leading to emergent epidemic behaviors such as localized outbreaks, percolation thresholds, and wave-like transmission patterns. These statistical effects capture the unpredictability of real epidemics, where some outbreaks die out early while others spread widely.
- Critical thresholds govern epidemic outcomes. The model exhibits phase transitions, where small changes in infection probability (P_infect) or recovery probability (P_recover) can determine whether an outbreak collapses or becomes

widespread. This aligns with percolation theory and epidemic models.

- Interventions such as social distancing and vaccination alter transmission dynamics. Reducing neighborhood size slows infection rates, while vaccination establishes immunity barriers, preventing further disease spread. The PCA framework effectively demonstrates how proactive and active interventions can influence epidemic trajectories.

Overall, this work highlights the strength of cellular automata models in capturing realistic epidemic dynamics that classical approaches often overlook. Future extensions, such as SEIR modeling, mobility effects, and dynamic intervention strategies, can further enhance the model's applicability to public health decision making and epidemic forecasting.

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